

# SESSION 3: ACCESSING EARTHSCOPE CLOUD STORE DATA WITH MSPASS


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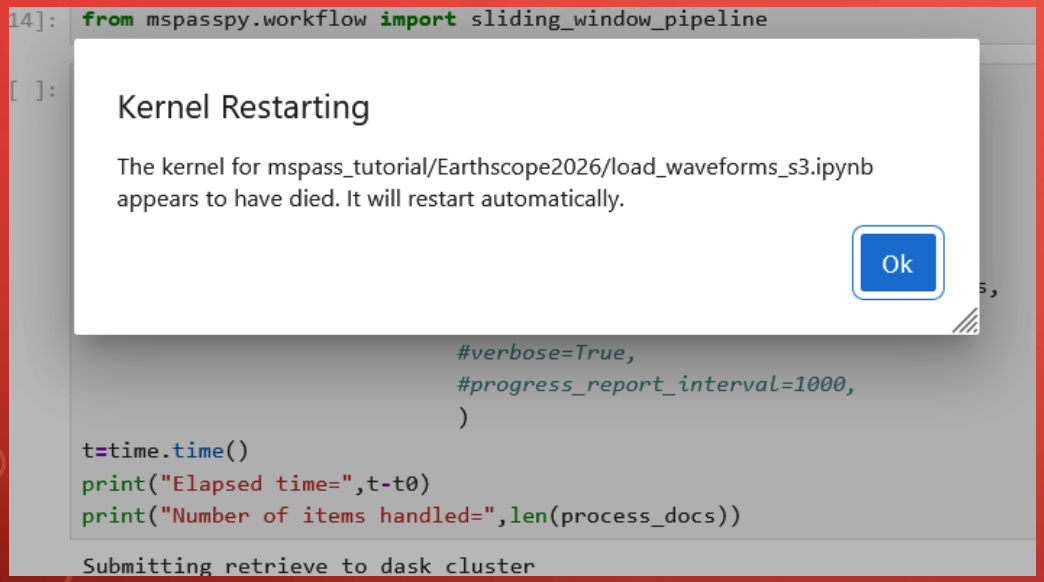
DEPT OF EARTH AND ATMOSPHERIC SCIENCES

INDIANA UNIVERSITY


MsPASS

Massive Parallel Analysis System for Seismologists

# QUESTIONS AND COMMENTS: PART 1 – RESPONSE BY ME




- Had a couple reports of this infamous (to us) jupyter splash screen
- What happens if the operating system kills the job
  - Usually means a memory fault
  - In this case it was hitting file quota limit on GeoLab
  - Happened when people had to rerun the load_waveforms_s3 notebook and didn't do a complete cleanup

# LESSONS

- If you get that infamous message, it means the OS killed the main python job script process
  - Memory fault
  - Something else that the OS considered a limit (in this case storage quota)
- Takes detective work to solve
  - Hypothesis driven testing
  - Shell command line tools to test file storage issues
  - Look at log files
    - I'll show you where they are on geolab for hints
    - They can be challenging to interpret
  - Use assorted memory monitoring tools – ask your favorite AI for suggestions

# OTHER QUESTIONS FROM HOWEWORK2

- What are the most common causes of metadata mismatches (such as net, sta , chan, and loc) that send data to the cementry collection, and how can they be prevented?
  - Usually not a problem with net, sta, chan, or loc but a mismatch of startime::endtime periods with IRIS holdings and what the provider gives as the time range of operation of a station – e.g. a maintenance window with a sensor swap.
  - The examples in this data  all  belonged in the cemetery – unreliable data best discarded
  - This is a type example of why error handlers and something like the cemetery are essential for large processing jobs to be stable

# ANY OTHERS?

# PART 1 OF TODAY:  IAN WANG ON USING MSPASS ON AN HPC CLUSTER

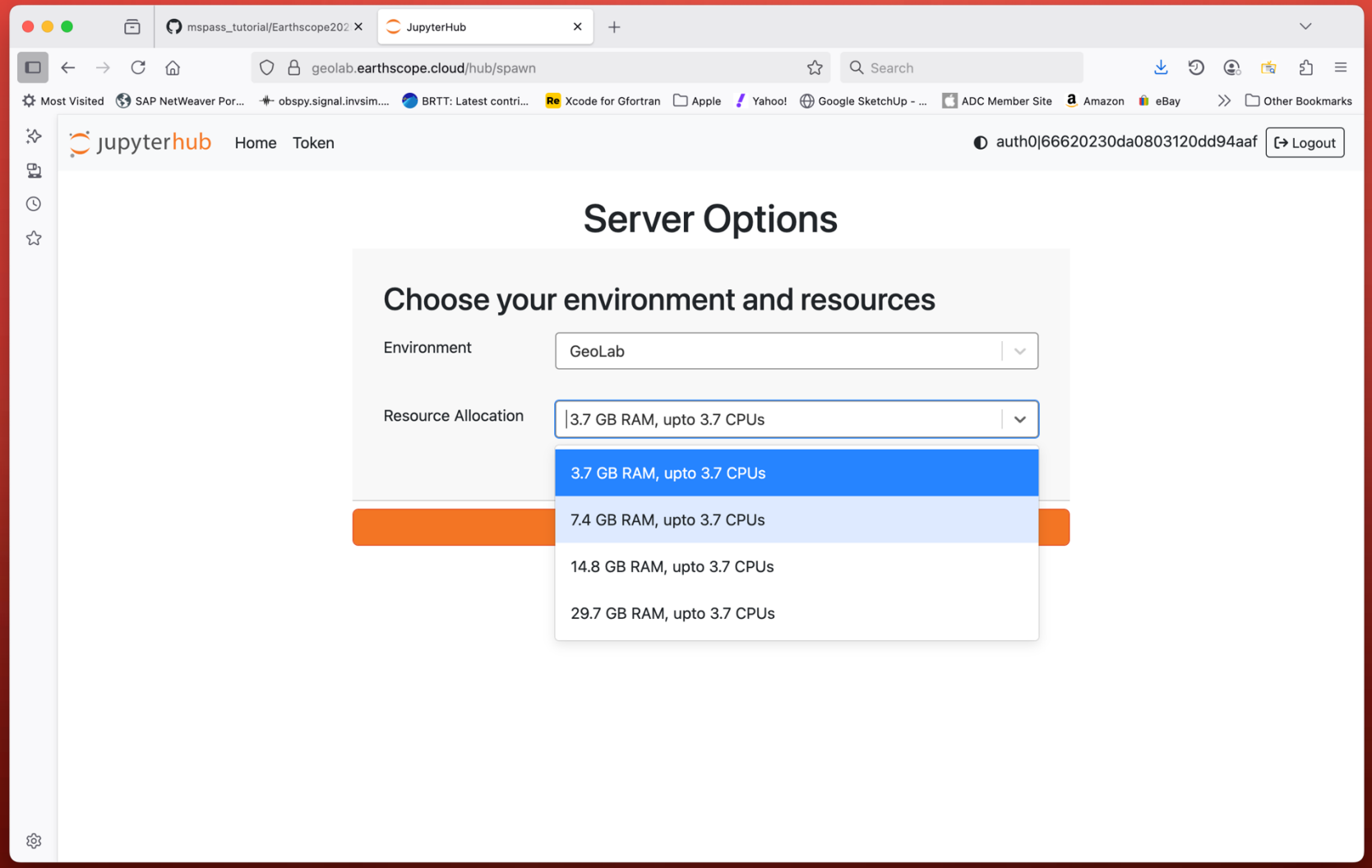
# CLOUD COMPUTING OVERVIEW

# MY TAKE AS A SCIENTIST ON CLOUD COMPUTING

- A lot of web sources are corrupted by commercial sales pitches
  - Technical audience
  - Nontechnical (manager) audience
- This is a huge subject with trillions of $ involved
- I will focus today on a few key concepts relevant to “big data” == MsPASS
  - Abstraction of what a processor is
  - Abstraction of data storage
  - Web view of data

# CONCEPT ONE: PROCESSOR ABSTRACTION

# WHAT IS THIS?


mspass_tutorial/Earthscope202
JupyterHub
geolab.earthscope.cloud/hub/spawn
Search
Most Visited
SAP NetWeaver Por...
obspy.signal.invsim....
BRTT: Latest contri...
Xcode for Gfortran
Apple
Yahoo!
Google SketchUp - ...
ADC Member Site
Amazon
eBay
Other Bookmarks
jupyterhub
Home
Token
auth0|66620230da0803120dd94aaf
Logout
Server Options
Choose your environment and resources
Environment
GeoLab
Resource Allocation
3.7 GB RAM, upto 3.7 CPUs
3.7 GB RAM, upto 3.7 CPUs
7.4 GB RAM, upto 3.7 CPUs
14.8 GB RAM, upto 3.7 CPUs
29.7 GB RAM, upto 3.7 CPUs

# KEY POINTS

- GeoLab is a cloud gateway
- Abstracts processors (“CPU” is not entirely accurate) and memory assigned to you while connected
- You have no clue what processor type or which node(s) of the cluster are assigned to your login

# HOW DOES THIS DIFFER FROM HPC




HPC



Amazon Data Center

# QUICK ANSWER

- Hardware is similar – multiple racks each with a stack of blades
- Software is completely different

# HPC VERSUS CLOUD SYSTEMS (A SCIENTIST PERSPECTIVE)

HPC Systems

- Created to support science
  - Support from NSF, DOE, and universities
  - Focus on simulation
- Grant support

- Set up to support academic research
- Simulation focus
  - Compute performance optimized
  - Batch processing (homogenous processing)

- Containers (virtual machines) are a newer application

Cloud Systems

- Driven by commercial interests
  - E-commerce market
  - Decreasing IT costs for the company
- TANSTAFL (There Ain't No Such Thing As a Free Lunch – Robert Heinlein)
- We are poorly financed customers
- Data handling and web focus
  - Optimized to handle large data volumes
  - Broad range of processing functionality
  - Center of e-commerce
- Entirely dependent upon containers

# HPC VERSUS CLOUD SOFTWARE

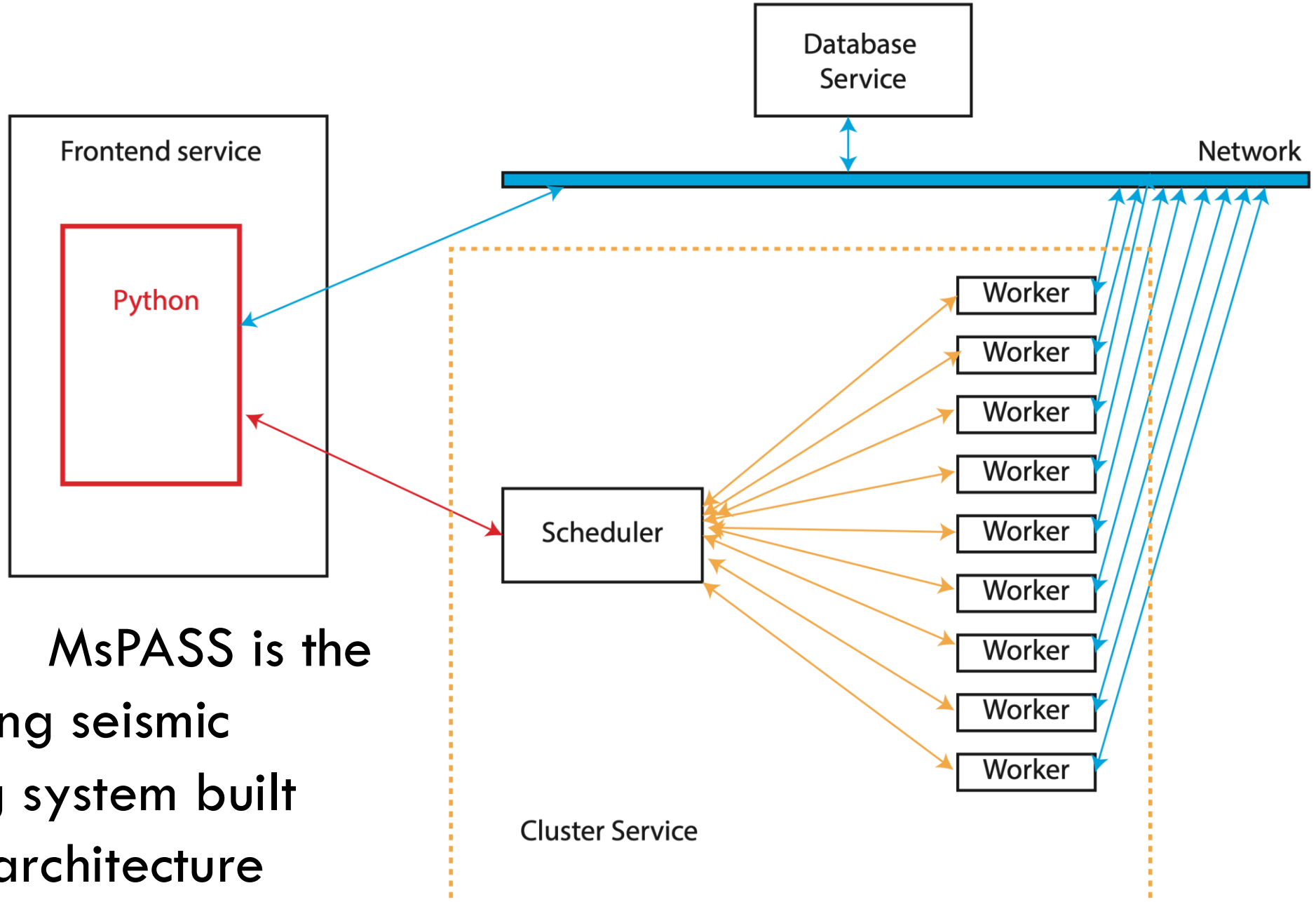
## HPC

- Batch processing
- Hardware conscious
  - Job requests resources as N nodes
  - Job requests other resources
- Components (processes) tightly connected together (mpi)

## Cloud

- Service abstraction
- Resources assigned dynamically
- Components (services) loosely connected (typically a url)

# MsPASS Services and Data Flow




Key point: MsPASS is the only existing seismic processing system built for cloud architecture

# SOME KEY POINTS:

- In a cloud system everything is abstracted
  - Processors
  - Memory use
  - Services
- HPC systems are “bare metal” – MsPASS creates the abstraction to make use indistinguishable from that on the cloud or a desktop

# CONCEPT TWO:  DATA ABSTRACTION

# DISCUSSION

- What is your concept of what a “computer file” is?
- Similarly, what is a “file system” like the unix or windows file system?

We will be using a zoom poll here

# COMPUTER FILE

- A collection of digital data stored on an external (to a running program) device (i.e. outside of system main memory)
- Loaded into system memory through a programming API
  - Loading of bits is a low-level operation done under the hood in any language
  - Usually “blocking”
    - Your program halts until IO operation completes
    - Operating system lets other process run while your program waits for input/output operation to complete
- Usually requires a special API to crack the format of the file after staged to memory
  - Miniseed
  - StationXML
  - etc

# FILE SYSTEM

- A system for organizing computer files
- Born from the then new technology of magnetic disks (random access storage)
- Traditional implementations tightly linked to disk storage
- Software file system emulators
  - Necessary evil to handle some hardware – notably large disk arrays
  - Used for decades – earliest IRIS servers used archival tape storage that simulated a unix file system

# AN EXAMPLE:  MODERN HPC CLUSTER FILE SYSTEMS

- Petabyte+ scale disk array found on most HPC clusters
- Some version of this was used at IRIS Data Center in Seattle that is now being phased out
- NOT a “unix file” system although the “path” (e.g. /N/scratch/pavlis) looks like a unix file system directory to a program
- In all cases I know where the data can actually be found is managed by a relational DBMS (File “Metadata”)

# WHY DID I INFLICT THAT ON YOU?

- Important to know that if you are working on a HPC system
  - Awful performance for large numbers of small files (e.g. SAC data)
  - You can crash an HPC cluster with excessive file operations
- Cloud systems
  - The actual data storage is similar to Lustre
    - binary blobs on giant disk arrays
    - Treated as “objects” instead of “files”
  - The API to access that data we will use (S3) does NOT simulate a file system
- Understanding blocking IO and it impacts on a parallel job is fundamental

# CLOUD STORAGE

- You likely use cloud stores every day with your phone and/or desktop
- Today's exercise focused on earthscope_sdk to access waveform archive now being moved to Amazon's cloud service called "S3"

# AMAZON S3 (SIMPLE STORAGE SERVICE)

- Best understood by examples we will run in a notebook after I finally stop talking
- Key concepts to take away from the exercise:
  - Standard python API is a package called boto3
  - S3 Bucket → File System
  - Object → File
  - A unique Object (File) is accessed by a unique key

# MSPASS DESIGN

- Was shaped by modern concepts in cloud computing
  - Service concepts (cluster, db, and frontend services)
  - Data abstraction (data objects and database reader - Session 1)
  - Containerized (distributed as docker container)
- Design allows the exact same package to run on
  - Desktop
  - HPC
  - Cloud (GeoLab)

# TO GEOLAB NOTEBOOK – SESSION3.IPYNB

# ISSUES WITH PROTOTYPE AND POSSIBLE SOLUTIONS

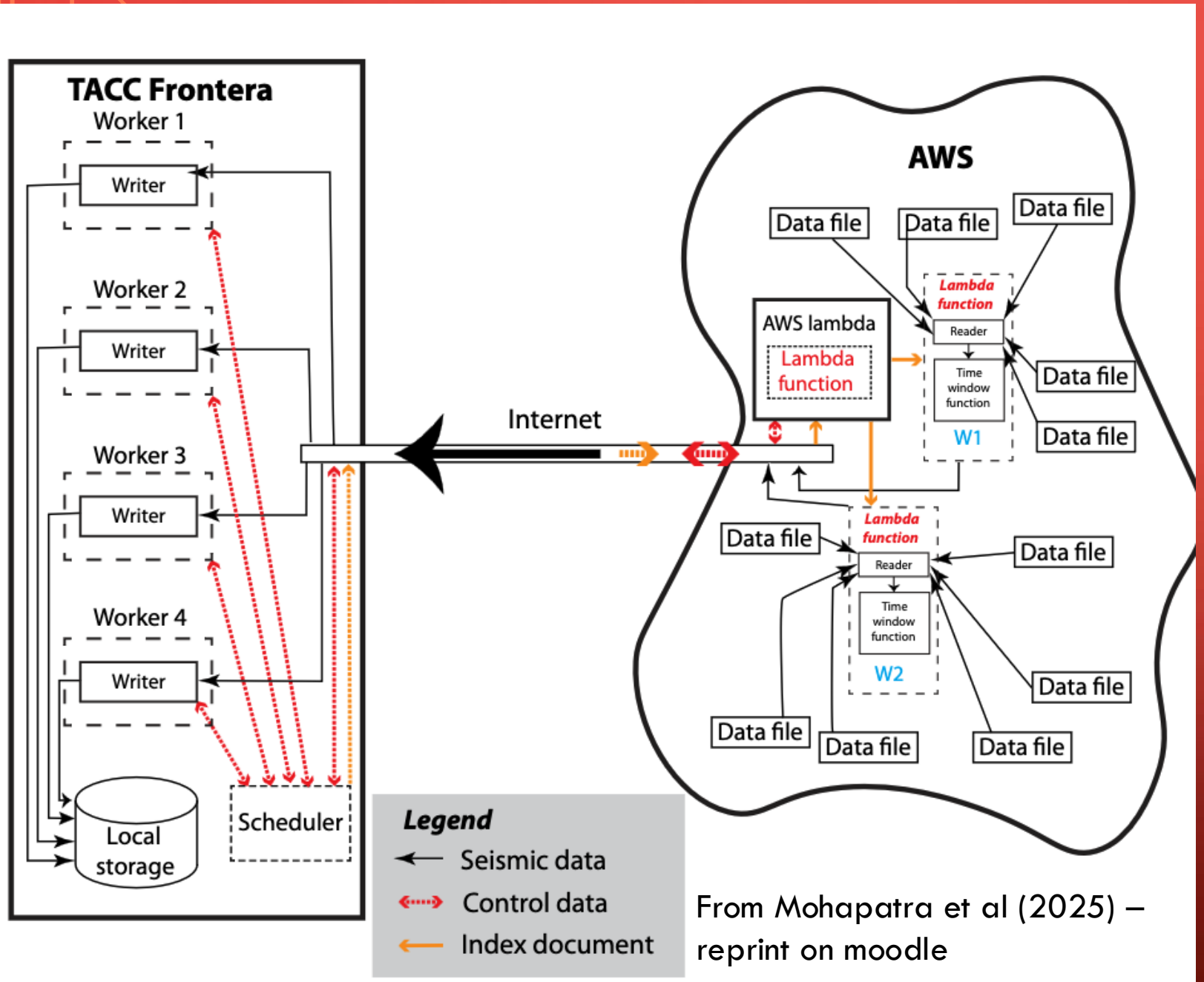
## Issues

- GeoLab storage limit
- Resource limit (GB-hour metric)

## Possible solutions

- Pay for your own S3 bucket for output (TANSTFL)
- Pay for additional processor time (TANSTFL)
- Set up lambda service
  - See our 2025 paper with link on Moodle
  - Very inexpensive (free currently possible) for modest performance

# Data Flow Diagram for using AWS Lambda service




From Mohapatra et al (2025) – reprint on moodle

Key points:

- The prototype you ran should work to drive a lambda
- You could run the left hand box on your local system

# EARTHSCOPE DATA SERVICES CLOUD PLANS

*CHAD TRABANT*

# CLOSING TOPICS

- Final Project:  Instructions on Moodle
  - Don't take on too much
    - Demonstrate independent application of MsPASS and/or
    - Demonstrate from words and coding you grasp one or more key concepts from the class
  - Defining problem and some prototyping is good enough for certificate
  - Plenty of time – Aug 15 deadline
  - I am willing help you continue after Aug 15 to take help you in your research
  - Use your favorite AI to help you – that is not only fair but recommended with MsPASS
  - Use email to Pavlis@iu.edu as I my monitoring of slack is sporatic after the class is over
- Certification deadline:  Aug 15 (complete all Moodle exercises by then)
- Cleanup your debris before leaving for last time:  GeoLab notebook file "Housecleaning.ipynb"

# DISCUSSION AND REVIEW

# OPTIONAL MATERIAL:  KEEP IN PRESENTATION BUT USED ONLY IF NEEDED FOR DISCUSSIONS

# MECHANICAL HARD DISK SPEEDS

| | | Metric → | IOPS | IOPS | MB/s | IOPS | IOPS | MB/s |
|---|---|---|---|---|---|---|---|---|
| Drive | | Block size → | 4 KB | 64 KB | 64 KB | 512 KB | 512 KB | large |
| Type | RPM | Access type → | random | random | random | random | random | sequential |
| SAS | 15 K | | 188 – 203 | 175 – 192 | 11.2 – 12.3 | 115 – 135 | 58.9 – 68.9 | 91.5 – 126.3 |
| FC | 15 K | | 163 – 178 | 151 – 169 | 9.7 – 10.8 | 97 – 123 | 49.7 – 63.1 | 73.5 – 127.5 |
| FC | 10 K | | 142 – 151 | 130 – 143 | 8.3 – 9.2 | 80 – 104 | 40.9 – 53.1 | 58.1 – 107.2 |
| SAS | 10 K | | 142 – 151 | 130 – 143 | 8.3 – 9.2 | 80 – 104 | 40.9 – 53.1 | 58.1 – 107.2 |
| SATA | 7200 | | 73 – 79 | 69 – 76 | 4.4 – 4.9 | 47 – 63 | 24.3 – 32.1 | 43.4 – 97.8 |
| SATA | 5400 | | 57 | 55 | 3.5 | 44 | 22.6 | |

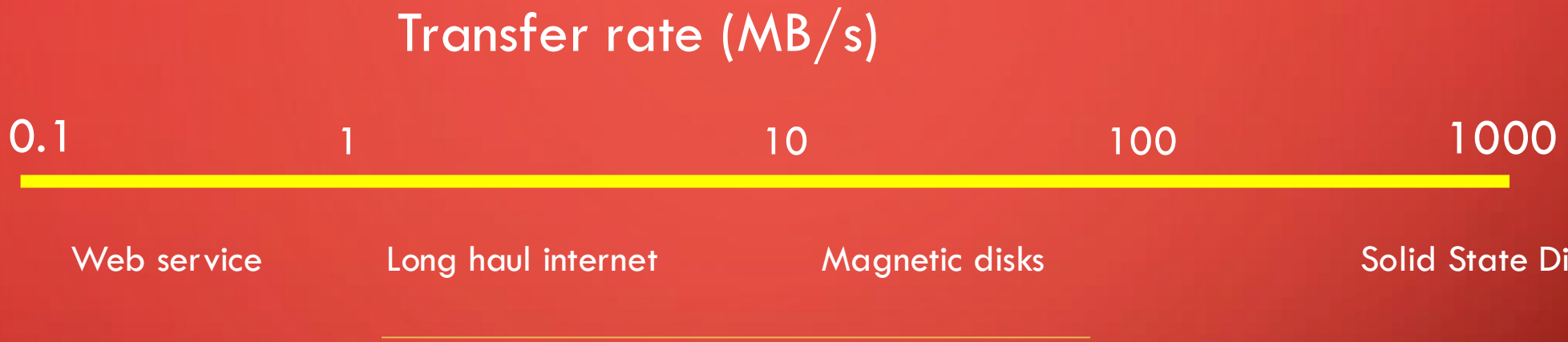
https://en.wikipedia.org/wiki/IOPS

# SOLID STATE DISK (SSD) SPEEDS

| Device | Type | IOPS | Interface | Notes |
|---|---|---|---|---|
| Intel X25-M G2 (MLC) | SSD | ~8,600 IOPS[12] | SATA 3 Gbit/s | Intel's data sheet[13] claims 6,600/8,600 IOPS (80 GB/160 GB version) and 35,000 IOPS for random 4 KB writes and reads, respectively. |
| Intel X25-E (SLC) | SSD | ~5,000 IOPS[14] | SATA 3 Gbit/s | Intel's data sheet[15] claims 3,300 IOPS and 35,000 IOPS for writes and reads, respectively. 5,000 IOPS are measured for a mix. Intel X25-E G1 has around 3 times higher IOPS compared to the Intel X25-M G2.[16] |
| G.Skill Phoenix Pro | SSD | ~20,000 IOPS[17] | SATA 3 Gbit/s | SandForce-1200 based SSD drives with enhanced firmware, states up to 50,000 IOPS, but benchmarking shows for this particular drive ~25,000 IOPS for random read and ~15,000 IOPS for random write.[17] |
| Samsung SSD 850 PRO | SSD | Up to[*neutrality is disputed*] 100,000 read IOPS<br>Up to[*neutrality is disputed*] 90,000 write IOPS[18][*non-primary source needed*] | SATA 6 Gbit/s | 4 KB aligned random I/O at QD32<br>10,000 read IOPS, 36,000 write IOPS at QD1<br>550 MB/s sequential read, 520 MB/s sequential write on 256 GB and larger models<br>550 MB/s sequential read, 470 MB/s sequential write on 128 GB model[18] |

https://en.wikipedia.org/wiki/IOPS

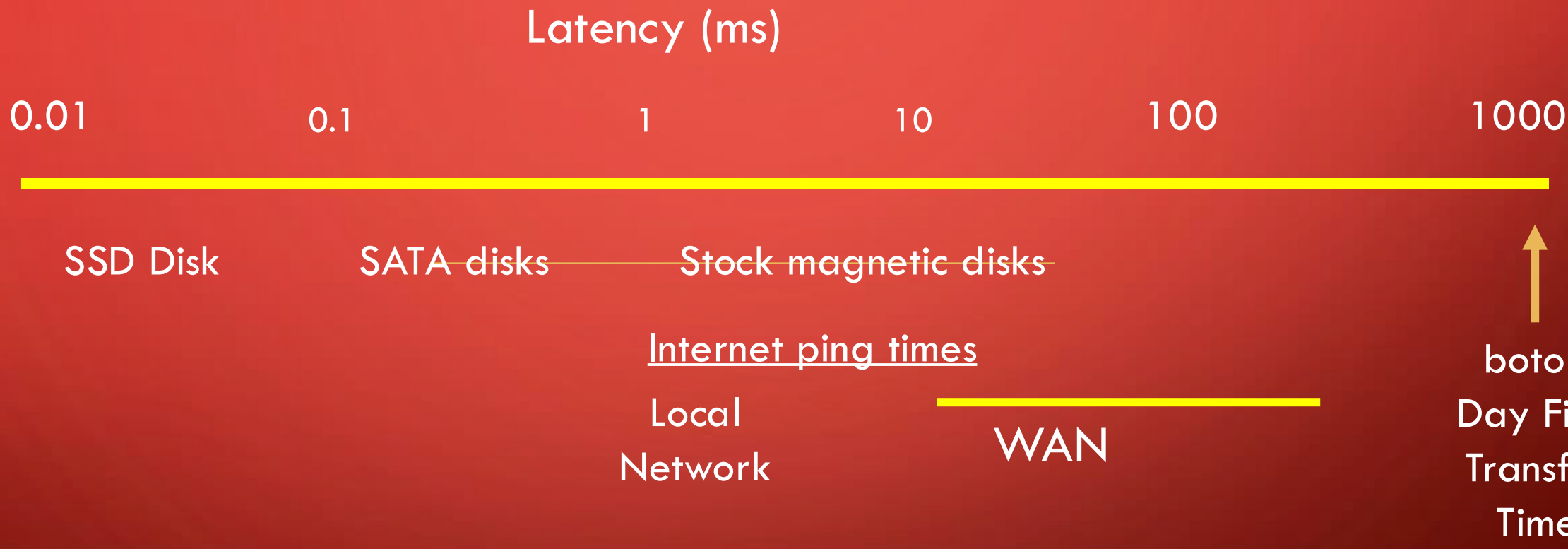
# Storage Device Transfer Speeds




Discussion points:

- human perception of time is not logarithmic
- Consider time to copy 1 TB ($10^{12}$) of data

# Latency




Useful source: https://planetscale.com/blog/io-devices-and-latency

# INPUT EXAMPLE COMPARISON

## Python Example

```
with open(fname, mode="rb") as fh:
    if foff > 0:
        fh.seek(foff)
    flh = io.BytesIO(fh.read(nbytes))
    st = obspy.read(flh, format=format)
```

## S3 boto3 equivalent

```
obj = s3client.get_object(Bucket=bucket,Key=s3key)
mseed_content=obj["Body"].read()
st = obspy.read(io.BytesIO(mseed_content),format="mseed")
```

Discussion:  What limits performance in both cases?

# KEY POINT YOU SHOULD TAKE FROM DISCUSSION

- With boto3 “get_object” requires moving binary data into memory
- Where that data lives matters
  - If data is local transfer speed can be gigabytes/s
  - If data is remote band limited by network (1 – 10+ Mb/s) – “long haul internet” problem